

Similarity and Distance

2026-05-06 · cheerful mango Haubentaucher

ENTER THE FEATURE SPACE

The Why

In supervised learning¹, the learning process relies on data samples x paired with their ground truth labels $\tilde{y}$, which allows a model θ to learn a mapping,

$$\begin{aligned}\theta(x) &= \hat{y}, \\ &\approx \tilde{y}.\end{aligned}$$

The presence of labels provides a clear signal for learning, as it allows optimization of a loss function that measures the discrepancy between predictions and true labels. The loss then guides the learning process, enabling the model to adjust its parameters along the gradient to minimize this discrepancy:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\hat{y}, \tilde{y}).$$

FACING THE ABSENCE OF LABELS, we are left with only the data samples² x and no explicit signal to guide learning. The methods we will explore in this course are designed to discover and leverage the inherent structure and patterns³ in the data itself, without relying on external supervision, thus mostly referred to as unsupervised learning.

IN A FIRST STEP this requires us to,

- define what we mean by pattern in the context of data.
- understand the geometry of data, which includes concepts of distance, similarity, and neighborhood.

¹ I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, 2016. <https://scholar.google.com/scholar?q=Goodfellow+Bengio+Courville+2016+deep+learning+book>

ON THE CONCEPT OF TRUTH, $\tilde{y}$, while being called ground truth, is often an approximation of the true label y . It may come from human annotation, measurements, or other sources.

² J. Rowley. The wisdom hierarchy: Representations of the dikw hierarchy. *Journal of Information and Communication Science*, 33(2):163–180, 2007. DOI: 10.1177/0165551506070706. <https://scholar.google.com/scholar?q=Rowley+2007+wisdom+hierarchy+DIKW>

³ C. M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006. ISBN 978-0387310732. <https://scholar.google.com/scholar?q=Bishop+2006+pattern+recognition+machine+learning>

KNOWLEDGE PYRAMID

Data (1) raw signal

→ Information (*banana*) endowed with meaning

→ Knowledge ($\neq$ *coffee*) structured and contextualized information

→ Wisdom or *knowing and appreciating the Why*.

- learn how to engineer data representations that make pattern more accessible.

THE FUNDAMENTAL QUESTION in unsupervised learning is to find pattern, represent it, and utilize it.

Hands On Experience

CONSIDER six data points in two dimensions. Which points seem to belong together? Why do you think that? How many groups do you see? To which group would you assign new points? Draw areas around each group.

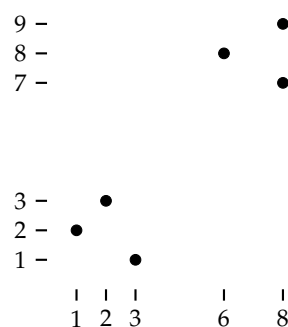


Figure 1: Six points in 2D. Can you see the two clusters? Or are there six? Or maybe one?

WITHOUT LABELS, pattern are ambiguous. With just a single additional point, the pattern can change drastically. Where does the new point belong? Does it belong to the first cluster, the second, or does it form its own cluster? Does this new point change the pattern of the original six points? How would you draw areas around the clusters now? How did that change your assumptions of the first pattern?

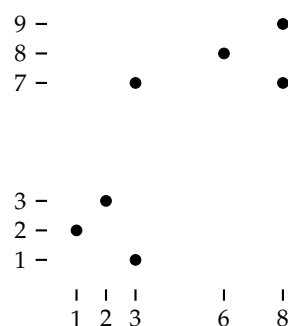


Figure 2: Seven points in 2D. The point at (3,7) creates ambiguity. Is it part of the first cluster, the second, or its own cluster?

HUMANS ARE REMARABLY GOOD AT FINDING PATTERN IN DATA, even with very little information, and even if it is completely off. It

PAREIDOLY is the tendency to perceive meaningful patterns in random data. Such as animals in clouds.

is a fundamental part of our cognition, and it is what allows us to make sense of the world. However, it also motivates the need for methods to quantify and formalize pattern. On that note it is also good practice to start with a hypothesis of what you expect to find in the data, and then test that hypothesis with the data, rather than just looking for any pattern that may emerge. And on a short managerial note, if the knowledge you will gain will not make you take action, then don't waste your time measuring it in the first place.

THE LEARNING OBJECTIVES of this lecture are that you will be able to:

- Understand the key role of distance and similarity in unsupervised learning and pattern recognition.
- Compute and interpret common distance metrics (Euclidean, Manhattan, Cosine, Mahalanobis).
- Apply appropriate normalization and scaling strategies and understand why they are necessary.

Distance and Similarity

QUITE INTUITIVELY, you already connected points and formed groups in terms of a similarity which was based on the closeness between data points. This concept will carry you very far.

A DISTANCE FUNCTION $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$, which means taking two samples out of the set X , mapping them to a non-negative real number, must satisfy:

$$\text{Non-negativity:} \quad d(x, y) \geq 0 \quad (1)$$

$$\text{Identity:} \quad d(x, y) = 0 \iff x = y \quad (2)$$

$$\text{Symmetry:} \quad d(x, y) = d(y, x) \quad (3)$$

$$\text{Triangle inequality:} \quad d(x, z) \leq d(x, y) + d(y, z) \quad (4)$$

EUCLIDEAN DISTANCE is probably the most familiar distance metric, measuring the straight-line distance between two points in space.

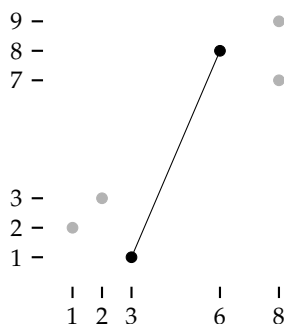


Figure 3: Euclidean distance between points $(3,1)$ and $(6,8)$.

And it was probably the one you implicitly used to connect the points in the previous exercise. It is defined as:

$$\begin{aligned} d_{\text{Euclidean}}(x, y) &= \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \\ &= \|x - y\|_2 \end{aligned}$$

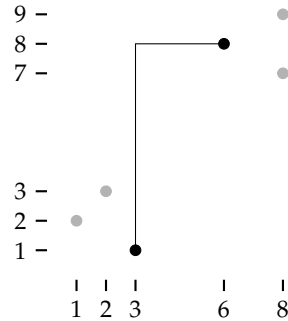
EUCLIDEAN DISTANCE assumes all dimensions are equally important and measured in the same units. It also assumes a continuous, smooth space. It is trivial to think of scenarios where these assumptions do not hold, such as when features have different scales (e.g. age vs. income) or when the features are not linearly related or not equally important (e.g. size vs. IQ). And then there are the cases where the units are ambiguous for the task like distances on maps.

Calculating the Euclidean distance between points $A = (3, 1)$ and $B = (6, 8)$:

$$\begin{aligned} d(A, B) &= \sqrt{(6-3)^2 + (8-1)^2} \\ &= \sqrt{9+49} \\ &= \sqrt{58} \approx 7.62 \end{aligned}$$

FEATURES can be thought of as dimensions in a vector space. Each data point is a vector in this space.

MANHATTAN DISTANCE is also called L_1 distance or taxicab distance. Instead of measuring the euclidean distance, it measures the axis-aligned path along grid lines:

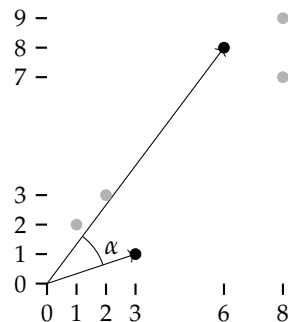


$$d_{\text{Manhattan}}(x, y) = \sum_{i=1}^n |x_i - y_i|$$

$$= \|x - y\|_1$$

ESPECIALLY USEFUL when movement is constrained to axes, such as for city blocks like the ones in Manhattan, pixel grids, or when you want robustness to outliers in individual dimensions.

COSINE SIMILARITY measures the angle between two vectors, ignoring their magnitude. It is especially useful when the correlation of the features of the data matters more than its absolute values.



When magnitude in the feature space does not matter only direction we use cosine similarity. It is defined as:

Figure 4: Manhattan (taxicab) distance between points (3,1) and (6,8): path follows grid lines.



Figure 5: Grid, 1811 (public domain) see Bridgeman Art Library v. Corel Corp.

Calculating the Manhattan distance between points $A = (3,1)$ and $B = (6,8)$:

$$d_M(A, B) = |6 - 3| + |8 - 1|$$

$$= 3 + 7$$

$$= 10$$

Figure 6: Cosine similarity between points (3,1) and (6,8): measuring the angle α between the position vectors.

Calculating the cosine similarity between points $A = (3,1)$ and $B = (6,8)$:

$$\text{sim}(A, B) = \frac{3 \cdot 6 + 1 \cdot 8}{\sqrt{3^2 + 1^2} \sqrt{6^2 + 8^2}}$$

$$= \frac{26}{\sqrt{10} \cdot 10}$$

$$\approx 0.82$$

$$\begin{aligned}
 d_{\text{cosine}}(x, y) &= 1 - \frac{x \cdot y}{\|x\| \|y\|} \\
 &= 1 - \frac{\sum_i x_i y_i}{\sqrt{\sum_i x_i^2} \sqrt{\sum_i y_i^2}}
 \end{aligned}$$

The right-hand term is the cosine similarity, which ranges from -1 (opposite) to $+1$ (identical direction), so the distance ranges from 0 to 2.

THINK of a vector multiplication as a projection of one vector onto another.

MAGNITUDE IS IRRELEVANT for example, in text analysis, documents can have vectors of very different lengths as some documents have more words than others, but cosine similarity will still focus on how many specific words per total word count. Similarly, in recommendation systems, users might have preference vectors with different scales, as some users rate more items, some less, but cosine similarity will compare the pattern of preferences.

MAHALANOBIS DISTANCE generalizes Euclidean distance by accounting for different variances and correlations among features. It measures distance in terms of the data's own covariance structure. In other words, it transforms the space so that distances are measured in units of standard deviation along principal axes, giving ellipsoidal rather than spherical neighborhoods. The covariance matrix Σ captures the spread and correlation of the data, and is defined as:

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_i)(x_i - \mu_i)^T$$

where μ is the mean vector of the data.

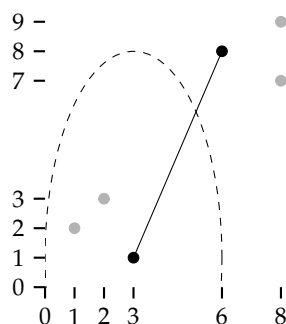


Figure 7: Mahalanobis distance between points (3,1) and (6,8): the dashed ellipse shows equidistant points under covariance Σ , replacing the Euclidean circle with an ellipse that reflects the data's spread.

Euclidean distance treats all dimensions equally, but real data often has different variances per dimension or correlations between

dimensions. The Mahalanobis distance compensates by weighting each dimension according to the data's spread. The inverse covariance matrix Σ^{-1} is used to weight the distance calculation, giving more weight to dimensions with less variance and less weight to dimensions with more variance.

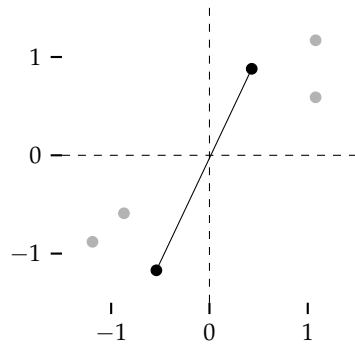
$$d_{\text{Mahalanobis}}(x, y) = \sqrt{(x - y)^T \Sigma^{-1} (x - y)}$$

where Σ is the covariance matrix of the data.

IT TRANSFORMS THE SPACE so that distances are measured in units of standard deviation along principal axes, giving ellipsoidal rather than spherical neighborhoods.

DATA PREPROCESSING, before computing distances, ensures that features are comparable. You have already seen, how the Mahalanobis distance incorporates the covariance structure of the data to account for different scales and correlations. With the Euclidean distance, if features are on different scales, the distance will be dominated by the larger-scaled feature. If one feature's measure has a different magnitude than the other, the larger-scaled feature will dominate the distance calculation. Preprocessing brings all features to a common ground by transforming the feature space into a standardized scale.

Z-SCORE NORMALIZATION transforms each feature to have mean 0 and standard deviation 1. This centers the data at the origin and scales each dimension by its own spread:



The z-score of each feature is computed as:

$$z_i = \frac{x_i - \mu_i}{\sigma_i} \quad (5)$$

where μ_i and σ_i are the mean and standard deviation of feature i .

Calculating the Mahalanobis distance between $A = (3, 1)$ and $B = (6, 8)$ with $\Sigma = \begin{pmatrix} 9 & 0 \\ 0 & 49 \end{pmatrix}$:

$$\begin{aligned} d_M(A, B) &= \sqrt{\frac{3^2}{9} + \frac{7^2}{49}} \\ &= \sqrt{1 + 1} \\ &= \sqrt{2} \\ &\approx 1.41 \end{aligned}$$

NORMALIZATION typically refers to scaling features to a specific range, such as $[0, 1]$. While standardization refers to centering features to have mean 0 and scaling to have standard deviation 1.

Figure 8: Our six points after z-score normalization: centered at the origin with unit variance in each dimension.

Normalizing point $A = (3, 1)$:

$$\begin{aligned} \mu &= (4.67, 5.00) \\ \sigma &= (3.08, 3.41) \end{aligned}$$

$$\begin{aligned} z_x &= \frac{3 - 4.67}{3.08} \\ &= -0.54 \end{aligned}$$

$$\begin{aligned} z_y &= \frac{1 - 5.00}{3.41} \\ &= -1.17 \end{aligned}$$

$$A' = (-0.54, -1.17)$$

USE WHEN features have different units or scales (e.g., age in years, vs. body height in millimeters). However, this is sensitive to outliers, as they influence μ and σ .

MIN-MAX SCALING transforms features to a fixed range $[0, 1]$. Each feature's minimum maps to 0 and its maximum to 1:

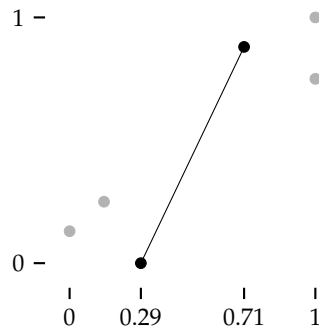


Figure 9: Our six points after min-max scaling: all values lie in $[0, 1]$.

$$x'_i = \frac{x_i - \min_i}{\max_i - \min_i} \quad (6)$$

USE WHEN you need bounded values. Unlike z-score, min-max guarantees a fixed range but is even more sensitive to outliers since a single extreme value stretches the scale and compresses other ranges.

CHOOSING A DISTANCE METRIC encodes your assumptions about what makes two data points similar in a given context. The same pair of points can appear close or far apart depending on the metric. The choice should be driven by the structure of your data, the context you are in, and the question you are answering.

A PRACTICAL RULE OF THUMB: try Euclidean first as a baseline, then ask whether the assumptions it makes, equal scales, independent features, magnitude matters, actually hold for your data. Each violation points toward a more suitable metric.

Scaling point $A = (3, 1)$ with $x \in [1, 8]$, $y \in [1, 9]$:

$$\begin{aligned} x' &= \frac{3 - 1}{8 - 1} \\ &= \frac{2}{7} \\ &\approx 0.29 \\ y' &= \frac{1 - 1}{9 - 1} \\ &= 0 \\ A' &= (0.29, 0) \end{aligned}$$

Distance between $A=(3, 1)$ and $B=(6, 8)$ under each metric:

$$\begin{aligned} d_{\text{Euclidean}} &= 7.62 \\ d_{\text{Manhattan}} &= 10 \\ d_{\text{Cosine}} &= 0.18 \\ d_{\text{Mahalanobis}} &= 1.41 \end{aligned}$$

Same points, four different answers. The distance defines the similarity.

Examples & Exercises

COMPUTING BY HAND builds the intuition that no amount of library calls can replace. Step through the arithmetic slowly.

GIVEN POINTS A , B , and C , compute the Euclidean distance $d_E(A, B)$, the Manhattan distance $d_M(A, C)$, and the cosine similarity $\text{sim}(A, B)$ with its corresponding cosine distance. If you feel you need more practice, go for it.

Euclidean distance sums squared coordinate differences:

$$\begin{aligned} d_E(A, B) &= \sqrt{(6-2)^2 + (1-5)^2 + (3-1)^2} \\ &= \sqrt{16 + 16 + 4} \\ &= \sqrt{36} \\ &= 6 \end{aligned}$$

Manhattan distance sums absolute differences instead, no squaring, no root:

$$\begin{aligned} d_M(A, C) &= |2-4| + |5-5| + |1-2| \\ &= 2 + 0 + 1 \\ &= 3 \end{aligned}$$

Cosine similarity measures the angle between the two vectors, independent of their length:

$$\begin{aligned} \text{sim}(A, B) &= \frac{A \cdot B}{\|A\| \|B\|} \\ &= \frac{2 \cdot 6 + 5 \cdot 1 + 1 \cdot 3}{\sqrt{4+25+1} \sqrt{36+1+9}} \\ &= \frac{20}{\sqrt{30} \sqrt{46}} \\ &\approx 0.54 \end{aligned}$$

The cosine distance follows as:

$$\begin{aligned} d_{\cos} &= 1 - 0.54 \\ &= 0.46 \end{aligned}$$

EXERCISES are for practice and reinforcing concepts. Try to solve them on your own first, try things, play with it, discuss, this is not a time trial. And there is no shame in not ending up at the right answer, in the same sense, that uncovering great questions and tossing them around is usually pretty fruitful on the long run.

$$A = (2, 5, 1)$$

$$B = (6, 1, 3)$$

$$C = (4, 5, 2)$$

THE METRIC IS A MODELLING CHOICE. There is no formula that tells you which distance to use, it follows from understanding your data and the context you are in. In the following scenarios, decide which distance metric is most appropriate and justify your choice. First

reason geometrically, then commit to an answer. Zoom-in so you can not see the margin notes, and try to solve it on your own first. Then read the margin notes for the answer.

A NEWS PLATFORM represents 10 000 articles as word-frequency vectors. Some articles are 100 words long, others 5 000 words. You want to find articles that cover similar topics.

A DELIVERY COMPANY plans routes through Manhattan's grid. Pickup locations are given as (street, avenue) coordinates.

A COFFEE QUALITY LAB measures sourness (0 to 200) and bitterness (0 to 500) for each batch.

GIVEN THREE COFFEE BATCHES measured by sourness and bitterness:

$$P_1 = (80, 400)$$

$$P_2 = (200, 160)$$

$$P_3 = (110, 220)$$

compute the sample covariance matrix and use it to find the Mahalanobis distance between P_1 and P_3 .

The mean vector:

$$\mu_x = 130$$

$$\mu_y = 260$$

$$\mu = (130, 260)$$

Deviations from the mean:

$$P_1 - \mu = (-50, 140)$$

$$P_2 - \mu = (70, -100)$$

$$P_3 - \mu = (-20, -40)$$

The sample covariance matrix sums all three outer products:

$$\begin{aligned}\Sigma &= \frac{1}{N-1} \sum (x_i - \mu)(x_i - \mu)^T \\ &= \frac{1}{2} \begin{pmatrix} 7800 & -13200 \\ -13200 & 31200 \end{pmatrix} \\ &= \begin{pmatrix} 3900 & -6600 \\ -6600 & 15600 \end{pmatrix}\end{aligned}$$

COSINE SIMILARITY. Article length inflates the magnitude of the frequency vector but does not change its direction. Cosine similarity ignores magnitude, comparing only the distribution of words.

MANHATTAN DISTANCE. Movement on a grid is constrained to horizontal and vertical steps. The shortest driving path between two intersections follows the L_1 norm, not the straight line.

MAHALANOBIS DISTANCE. The features live on different scales and are correlated. Mahalanobis accounts for both by using the covariance matrix, producing ellipsoidal contours that align with the data's spread.

We invert Σ analytically using the 2×2 formula:

$$\begin{aligned}\det(\Sigma) &= 3900 \cdot 15600 - 6600^2 \\ &= 17,280,000 \\ \Sigma^{-1} &= \frac{1}{17,280,000} \begin{pmatrix} 15600 & 6600 \\ 6600 & 3900 \end{pmatrix}\end{aligned}$$

The difference vector is $(x - y) = P_3 - P_1 = (30, -180)$. Apply Σ^{-1} :

$$\begin{aligned}\Sigma^{-1}(x - y) &= \frac{1}{17,280,000} \begin{pmatrix} 15600 & 6600 \\ 6600 & 3900 \end{pmatrix} \begin{pmatrix} 30 \\ -180 \end{pmatrix} \\ &= \frac{1}{17,280,000} \begin{pmatrix} -720,000 \\ -504,000 \end{pmatrix}\end{aligned}$$

Taking the dot product with $(x - y)$ gives the squared distance:

$$\begin{aligned}d_M^2 &= (x - y)^T (\Sigma^{-1}(x - y)) \\ &= \frac{1}{17,280,000} (30, -180) \begin{pmatrix} -720,000 \\ -504,000 \end{pmatrix} \\ &= \frac{69,120,000}{17,280,000} \\ &= 4 \\ d_M &= \sqrt{4} \\ &= 2\end{aligned}$$

Compare with the Euclidean distance, which does not account for the different scales and correlation. The raw displacement of 30 sourness units and 180 bitterness units yields an overall distance of 182.5, making P_3 seem very far from P_1 , driven almost entirely by the bitterness difference:

$$\begin{aligned}d_E(P_1, P_3) &= \sqrt{(110 - 80)^2 + (220 - 400)^2} \\ &= \sqrt{900 + 32400} \\ &= \sqrt{33300} \\ &\approx 182.5\end{aligned}$$

The Mahalanobis distance is far smaller because the covariance matrix rescales for the large variance difference between features. The bitterness axis has a much larger spread ($\sigma^2=15600$ vs $\sigma^2=3900$), so a displacement of 180 bitterness units is less unusual than it appears, and contributes less to the Mahalanobis distance accordingly.

For a 2×2 matrix:

$$\begin{aligned}A &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ A^{-1} &= \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\end{aligned}$$

Swap the diagonal, negate the off-diagonal, divide by the determinant.

INTUITIVELY, A^{-1} undoes the transformation A : if A stretches and rotates space, A^{-1} maps everything back, with $AA^{-1} = I$. For Σ , the inverse rescales each dimension by its variance and removes correlations, so distances are measured in standard-deviation units.

THE PARENTHESES in $(x - y)^T (\Sigma^{-1}(x - y))$ highlight that $\Sigma^{-1}(x - y)$ is computed as one step. In this exercise we form Σ^{-1} explicitly because the matrix is 2×2 . In higher dimensions, computing the full inverse is expensive and amplifies rounding errors. Instead, you solve $\Sigma z = (x - y)$ for z , which gives the same result as $\Sigma^{-1}(x - y)$ without ever forming the inverse — just as you can compute $5/3$ by dividing directly rather than first finding $1/3$ and then multiplying by 5.

STANDARDIZATION does not change the data, it changes the expanse and spread of the space you measure it in. Units disappear; and make way for a more abstract notion of distance, in terms of how many standard deviations apart the data points are.

THE EFFECT OF STANDARDIZATION on café data. Three cafés C_1 , C_2 , C_3 recorded as daily cups sold and customer rating.

Compute the Euclidean distance on raw features, standardize all three points, and recompute. On raw features, daily sales dominate completely; the two pairs involving C_2 look equally close:

$$\begin{aligned} d_E(C_1, C_2) &= \sqrt{(500-300)^2 + (6-7)^2} \\ &= \sqrt{40\,000 + 1} \\ &\approx 200.0 \\ d_E(C_2, C_3) &= \sqrt{(700-500)^2 + (3-6)^2} \\ &= \sqrt{40\,000 + 9} \\ &\approx 200.0 \end{aligned}$$

The rating differences (1 and 9) are drowned out by the cup counts (40,000 each). The means and standard deviations are:

$$\begin{aligned} \mu_{\text{cups}} &= 500, & \sigma_{\text{cups}} &= 200 \\ \mu_{\text{rating}} &\approx 5.33, & \sigma_{\text{rating}} &\approx 2.08 \end{aligned}$$

After z-score normalization the cafés map to:

$$C'_1 \approx (-1, 0.80), \quad C'_2 \approx (0, 0.32), \quad C'_3 \approx (1, -1.12)$$

Recomputing the distances:

$$\begin{aligned} d(C'_1, C'_2) &= \sqrt{(0-(-1))^2 + (0.32-0.80)^2} \\ &= \sqrt{1 + 0.23} \\ &\approx 1.11 \\ d(C'_2, C'_3) &= \sqrt{(1-0)^2 + (-1.12-0.32)^2} \\ &= \sqrt{1 + 2.07} \\ &\approx 1.75 \end{aligned}$$

C_2 is now clearly closer to C_1 than to C_3 . In raw space, sales made both pairs look identical; after standardization, the similar reviews of C_1 and C_2 pull them together while C_3 's low rating pushes it away.

A TOY DATASET TO EXPLORE. You are given measurements of 20 coffee samples with three features: *acidity* (pH, 4.0 to 7.0), *bitterness*

$$C_1 = (300, 7)$$

$$C_2 = (500, 6)$$

$$C_3 = (700, 3)$$

daily cups sold, customer rating (1 to 10)

THE CUPS standardize cleanly because 300, 500, 700 are evenly spaced. The ratings do not: 7, 6, 3 give a non-integer mean and an irrational standard deviation. This is what real data looks like.

CODE will be provided as a Python notebook. Use it as a starting point, break things, and observe what changes.

20 coffee samples

acidity (pH, 4.0 to 7.0)

bitterness (1 to 10)

brew strength (mg/mL, 5 to 25)

four varieties: cold brew, drip, espresso, latte

(1 to 10), and *brew strength* (mg/mL, 5 to 25). The samples belong to four varieties: cold brew, drip, espresso, and latte. Load the dataset and scatter-plot acidity vs. bitterness. Then compute the Euclidean distance matrix and identify the most similar pair, is the result plausible? Apply z-score normalization, recompute the distance matrix, and observe how the nearest neighbors shift. Compute cosine similarity between all pairs and compare it to the Euclidean grouping. Finally, compute the sample covariance matrix of the normalized data and check whether any features are correlated.

WHAT TO OBSERVE. Without normalization, brew strength dominates similarity because its scale (5 to 25 mg/mL) dwarfs the others. A latte and an espresso with similar strength appear closer than two espressos with the same acidity and bitterness profile. After normalization, all three features contribute equally and the four natural clusters become visible. Cosine similarity groups samples by the *ratio* of their features, which may merge varieties with similar flavor profiles but very different intensities.

Self-Reflection and Recap

SELF-REFLECTION Questions which can guide your thoughts during the exercises and afterwards:

- How do we distinguish between supervised and unsupervised learning?
- Elaborate on the differences between $\tilde{y}$, $\hat{y}$, and y ?
- What are the key properties that define a distance metric?
- How do different distance metrics capture different notions of similarity?
- What is a feature space, and how do data points relate to it?
- What do we mean by feature engineering and feature transformation?
- What can happen when features are on different scales or magnitudes?
- How can we engineer and transform feature spaces?
- Why do we transform feature spaces with respect to pattern? And what are the common transformations?

RECAP of Key Concepts:

- Unsupervised learning discovers pattern in unlabeled data
- Distance metrics define what similar means
- Feature spaces can be engineered and transformed to reveal different patterns

SO FAR, WE DID MEASURE SIMILARITY BETWEEN SINGLE DATA POINTS. How do we embed this information across data points? Is it possible to represent pattern in the data as a whole, rather than just pairwise similarities? In the next chapter we explore clusters and clustering algorithms, which are the most classic way to represent pattern in unsupervised learning.

WITHOUT KNOWING ABOUT LABELS OR SEMANTICS we can tell which samples are similar.

TEASER. How do we represent structure beyond pairwise similarities and distances?

FEEDBACK